

- **Radix UI:** An open-source component library providing unstyled, accessible primitives for building high-quality design systems. Radix UI handles complex interactions such as dialogs, dropdowns, and tooltips with proper ARIA attributes.
- **Razorpay:** India's leading payment gateway, integrated for processing waste transactions. Razorpay supports UPI, credit/debit cards, net banking, and wallet payments with PCI DSS Level 1 compliance.
- **Google Genkit AI:** A framework for building AI-powered features. AgriTrace uses Genkit for intelligent waste classification, helping farmers identify waste types through AI-assisted categorization.
- **Cloudinary:** A cloud-based media management service used for uploading, storing, and optimizing waste report photographs with automatic image compression and format conversion.

1.4 Organization of the Report

This report is organized into fourteen chapters, each covering a specific aspect of the project:

- **Chapter 1 – Introduction:** Provides the background, aim, motivation, and technology overview of the AgriTrace project.
- **Chapter 2 – Problem Statement:** Defines the problem being addressed with supporting statistics and analysis of existing solutions.
- **Chapter 3 – Literature Survey:** Reviews existing research and systems related to agricultural waste management, digital platforms, and the technologies used.
- **Chapter 4 – Objectives and Scope:** Lists the specific objectives and defines the boundaries and scope of the project.
- **Chapter 5 – Requirement Analysis:** Documents the hardware and software requirements along with functional and non-functional requirements.
- **Chapter 6 – Methodology:** Describes the Agile Scrum development methodology adopted, including sprint planning and development tools.
- **Chapter 7 – System Design and Modeling:** Presents the system's design through Gantt charts, data flow diagrams, use case diagrams, activity diagrams, ER diagrams, and other UML models.
- **Chapter 8 – Coding:** Contains the key source code modules with detailed explanations of backend services, authentication, location services, notifications, payments, and UI components.